

1. Introduction to Digital Image Processing (4 hrs)

1.1. The Origin of Digital Image Processing:

- The pictorial representation of an object is known as formation of an image.
- A digital image is a 2D light intensity function $f(x, y)$ where 'x' and 'y' denotes spatial coordinates and value of the function "f" at any point (x, y) is proportional to the brightness (gray level) of an image at that point (x, y) .
- A digital image is a representation of a real image as a set of numbers that can be stored and handled by a digital computer.
- Digital image processing is the processing of digital image by using digital computer on the basis of digitization (sampling & Quantization).

Digital Image Representation:

- A digital image is an image $f(x, y)$ that has been discretized both in spatial coordinates and in brightness.
- It is represented by 2D integer array.
- The digitized brightness values is ~~gr~~ called gray level.
- Digital image can be represented as a matrix shown below:

$$f(x, y) = \begin{bmatrix} f(0, 0), f(1, 0) & \dots & \dots & \dots \\ f(0, 1) & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & f(x_{\max}, y_{\max}) \end{bmatrix}$$

- Here, the rows and columns identify the point in the image and the corresponding matrix element values identify the gray level at that point.
- The element in digital image is called pixel or pels or picture point.

1.2. Examples of Fields that use Digital Image Processing:

→ Following are the fields where digital image processing are used:

i> Office Automation:

→ In office automation, digital image processing are used for OCR (Optical Character Reader), document processing etc.

ii> Industrial Automation:

→ In industrial automation, DIP is used for Robotics, Automatic Assembly etc.

iii> Bio-Medical Automation:

→ In Bio-medical automation, DIP is used for ECG, EEG, X-Ray etc.

iv> Information Technology (IT):

→ In information technology, DIP is used for fax, video conferencing etc.

v> Entertainment:

→ In entertainment, DIP is used for video game, Visual Animations etc.

vi> Military Application:

→ In military application, DIP is used for target identification and missile guidance and detection.

iv> Criminology Security:

→ In criminology, DIP is used for bio-metric detection such as fingerprint detection, face detection, retina detection etc.

1.3. Fundamental Steps in Image Processing:

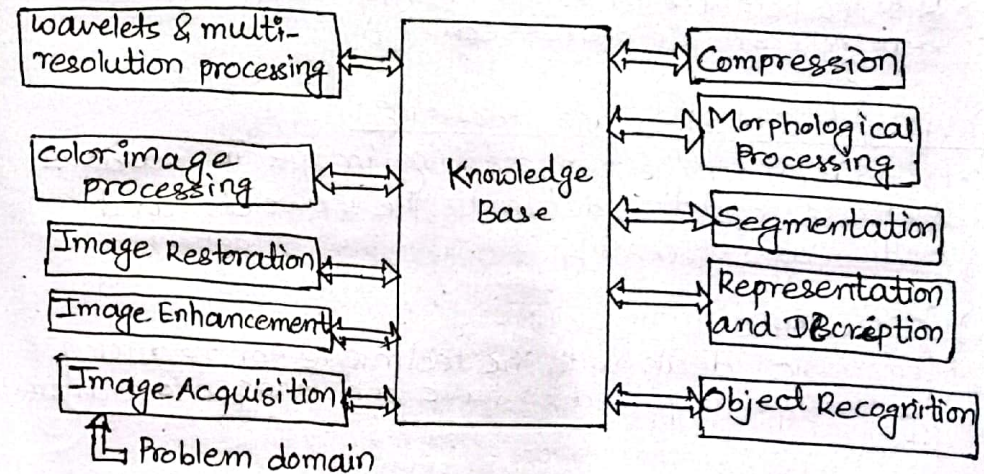


Fig. Steps in Digital Image Processing

i> Image Acquisition:

→ Image acquisition is the first process in which pre-processing activities like scaling and rotating are done.

ii> Image Enhancement:

→ In image enhancement, the original image is processed in such a manner that the resultant image will be more suitable with compared to original image.

iii) Image Restoration:

- Image restoration is an area that deals with improving appearance of an image.
- Basically, image enhancement is a subjective process but image restoration is an objective process.

iv) Color Image Processing:

- With the help of color, object identification and extraction from the scene is processed.

v) Wavelets & Multi-Image Processing:

- Foundation for the representing image in various degree of resolution deals to the wavelets and multi-image processing.

vi) Compression:

- Compression deals with the technique for reducing the storage required to save an image to transmit it.

vii) Morphological Processing:

- Morphological processing deals with tools for extraction of an image component that are useful in representation and description of shape.

viii) Segmentation:

- Segmentation deals with the partition of an image into different parts and objectives.

ix) Representation:

- After an image has been segmented into sub-regions then the resultant segmented pixels are represented and described in a suitable form for further analysis.

x) Object Recognition:

- Object recognition processes that the level to an object based on its descriptors.

1.4. Elements of Digital Image Processing Systems:

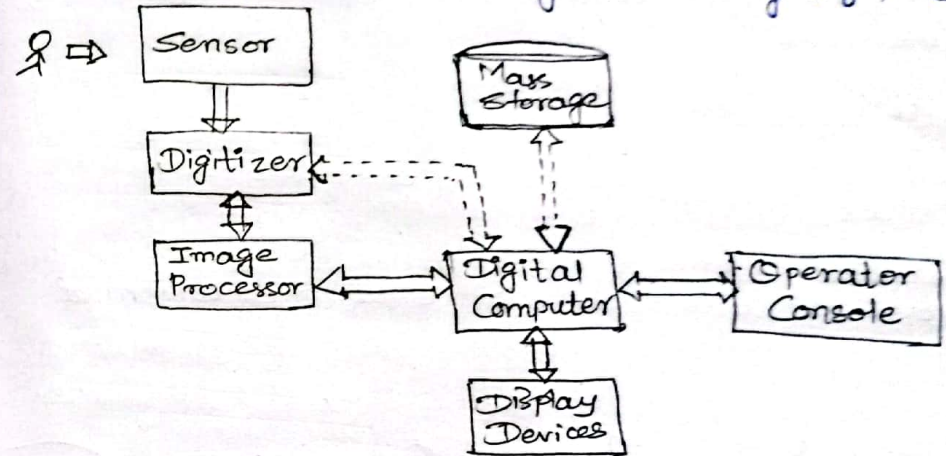


Fig. Elements of DIP.

i) Sensor:

- Sensor intercepts the energy propagating from the scene and transform it to produce an intensity image.

ii) Digitizer:

- Digitizer converts an intensity image into numerical representation which is suitable for input into a digital computer.

iii) Image Processors:

- Image Processor interfaces with digital computer to provide easy of programming for this purpose we can use micro-computer or mainframe computer.

iv> Digital Computer:

→ It always digitize the required image by the help of image pre-processor.

v> Mass Storage:

→ Mass storage is used to store digital image line by line.

vi> Display devices:

→ Display device produces and shows a visual of numerical values stored in a computer.

vii> Operator Console:

→ Operator console controls the full mechanism of image processing system.

1.5. A Typical Image Processing Sequence:

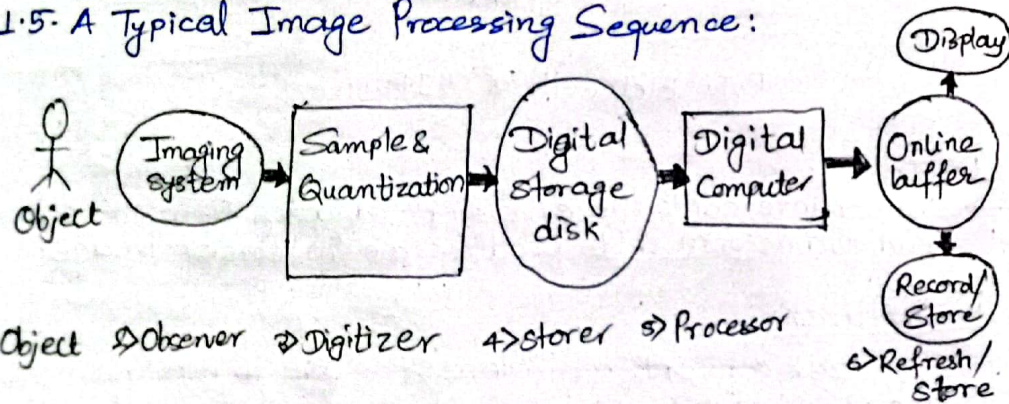


Fig. Image Processing Sequence

In typical im

→ In typical image processing sequence involves various components. The image of an object travel through various blocks.

i> Firstly, the object is placed and sensed by the system and image is formed.

ii> This image is digitized by the process of sampling & quantization.

iii> Now the image is in the digital form.

iv> A digital image is stored in digital storage system.

v> The digital computer is connected to digital storage disk where all the manipulation of digital image is done.

vi> The online buffer is available where the information are stored as a memory.

vii> The image is displayed through the display block which may be monitor, TV etc.

viii> Finally, the image is recorded in a recording block.

But sometimes some problems may occur when the image is processed sequentially into digital computer.

1.6. Image Sampling and Quantization:

→ This includes the first elements as a digitization.

i) [Digitization]:

→ A physical image is changed into digital image by performing sampling and Quantization which is known as digitization.

ii) [Sampling]:

→ Sampling is the process of taking measurements at regular space intervals.

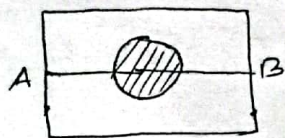
→ Sampling is also digitizing the spatial co-ordinates values (x, y) .

→ Sampling can be defined as a selection of discrete location in the continuous 2D space.

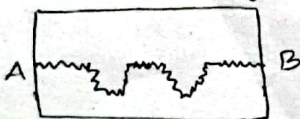
iii) [Quantization]:

→ Quantization is the process of mapping the measured intensity to one of the finite numbers of discrete levels.

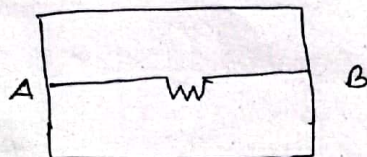
→ Quantization is also digitizing the amplitude values.



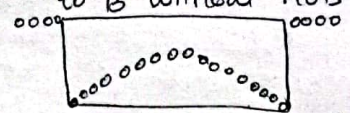
1) Continuous Type



2) Scanline A to B in image with noise



3) Ideal scanline A to B without noise



4) Sampling & Quantization

1.7. Some Basic relationship like Neighbors, Connectivity, Distance measures between pixels.

i) Neighbors of pixel:

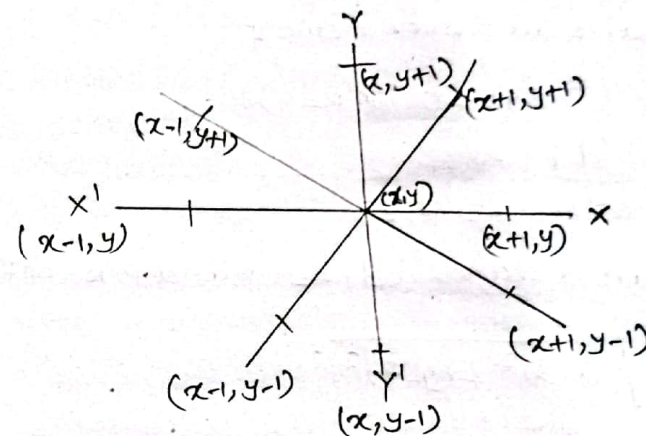


Fig. Neighbors of pixel

→ A pixel at co-ordinate (x, y) has a four horizontal and four vertical neighbors whose co-ordinates are given by: $(x+1, y+1)$; $(x-1, y-1)$; $(x-1, y+1)$; $(x+1, y-1)$; $(x, y+1)$; $(x, y-1)$; $(x+1, y)$; $(x-1, y)$.

→ Each pixel is at unit distance from (x, y) and called as $N_8(P)$.

ii) Distance Measures:

→ For pixels p, q and z with respective co-ordinates $(x, y), (s, t)$ and (v, w) .

→ D is the distance function or metric if

$$a) D(p, q) \geq 0 \quad (D(p, q) = 0 \text{ iff } p = q)$$

$$b) D(p, q) = D(q, p) \text{ \& }$$

$$c) D(p, z) \leq D(p, q) + D(q, z)$$

→ The Euclidean distance between p and q is defined as

$$D_e(p, q) = \sqrt{(x-s)^2 + (y-t)^2}$$

→ The distance D_4 (also called city-block distance) between p and q is defined as

$$D_4(p, q) = |x-s| + |y-t|$$

- The pixels with $D_4 = 1$ are the 4-neighbors of (x, y) .

→ The distance D_8 (also called chessboard distance) between p and q is defined as

$$D_8(p, q) = \max(|x-s|, |y-t|)$$

- The pixels with $D_8 = 1$ are the 8-neighbors of (x, y) .

iii) Connectivity between Pixels:

→ Two pixels are said to be connected if they are adjacent in some sense.

→ There are four three types of connectivity on the basis of adjacency.

i) 4-Connectivity:

→ Two or more pixels are said to be 4-connected if they are 4-adjacent with each other's.

ii) 8-connectivity:

→ Two or more pixels are said to be 8-connected if they are 8-adjacent with each other's.

iii) m-connectivity:

→ Two or more pixels are said to be m-connected if they are m-adjacent with each other's.

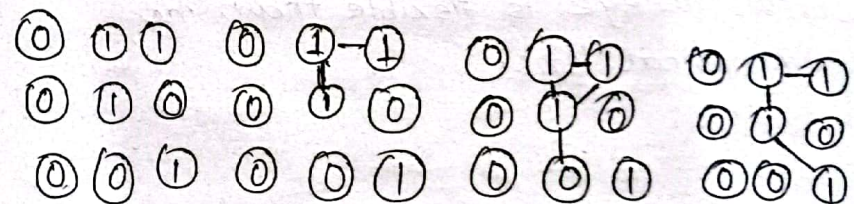


Fig. An arrangement of pixels Fig. 4-connectivity of pixels Fig. 8-connectivity of pixels Fig. m-connectivity of pixels

1.8 Elements of Visual Perception:

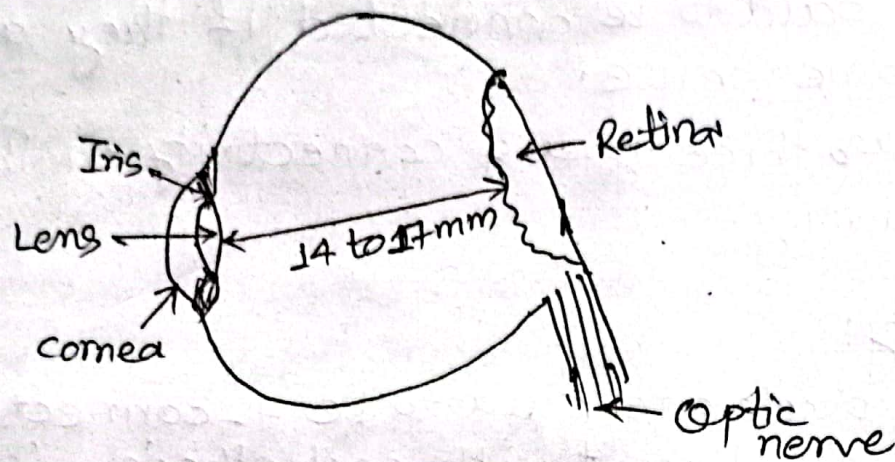


Fig. Cross Section of Human Eye

- Image analysis plays central role in the image processing.
- Therefore, the study of human visual perception is important which is represented as cross-section of human eye is as shown above.
- The human eye nearly a sphere with an average diam diameter of 14 mm to 17 mm.
- The lens of the eye is flexible than the ordinary optical lens.

2. Image Enhancement in Spatial Domain (7-hrs)

2.1. Some basic Gray Level Transformations:

- Image enhancement refers to the sharpening or to improve the quality of an image by enhancement of pixel or point in a spatial domain.
- Some basic gray level transformations are needed to enhance the image in a spatial domain.
 - i> Point operations
 - ii> Contrast stretching
 - iii> Thresholding
 - iv> Digital Negative
 - v> Intensity level slicing
 - vi> Bit Plane Slicing

2.1.1. Point Operations:

- Point operation is defined as a function that are performed on each pixel of an image.
- Point operation is unary point operation if single image is modified.
- Point operation is binary point operation if two images combine in the same manner.
- Point operation is ternary point operation if three images combine in the same manner.

→ Unary operation can be defined as

$$T_{xy} = \text{function}(P(x,y))$$

where,

T_{xy} = Transformed pixel value

function = Numeric transform

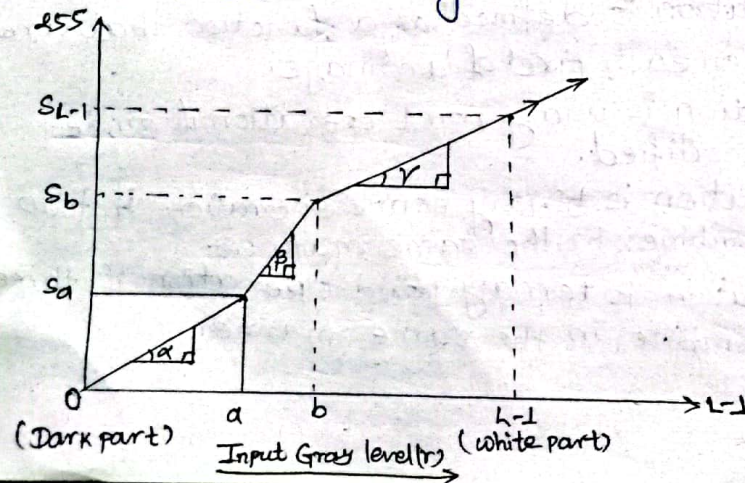
$P(x,y)$ = Input Pixel Value

→ This operation can be used in many situations and different purposes such as: contrast stretching, noise clipping, window slicing and histogram modeling.

→ So, in the point operation, a given gray level, $u \in [0, L-1]$ is mapped into a gray level $v \in [0, L-1]$, according to a transformation.

$$\text{i.e. } v = f(u) \quad \left\{ \begin{array}{l} L-1 = 255 \\ L = \text{max. gray level} \end{array} \right\}$$

2.1.2. Contrast stretching:



→ In many times, we obtain low contrast image due to poor illumination or due to the wrong setting of lens aperture.

→ The idea behind contrast stretching is to increase the dynamic range of gray level in the image.

→ The main idea is to increase the contrast of the image by making dark portion more darker & bright portion more brighter.

→ To obtain the contrast stretching, we should have some formulations as below:

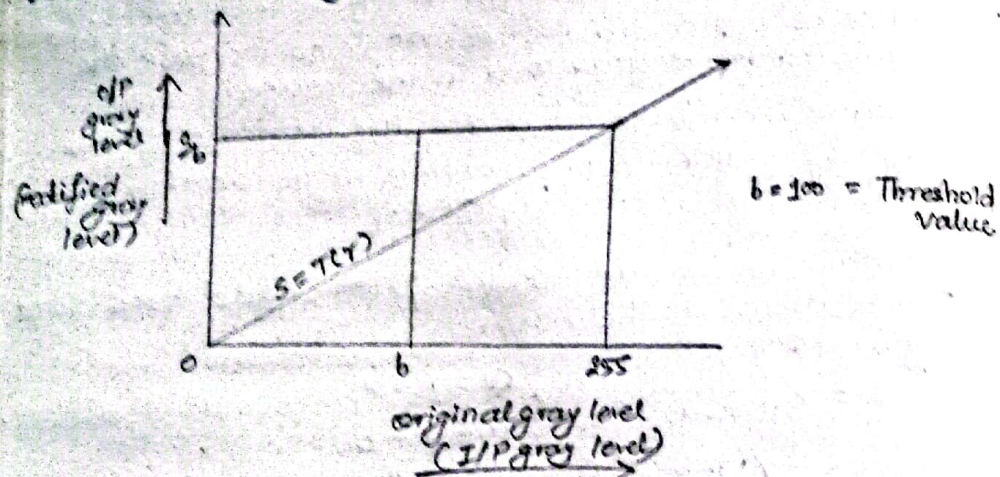
$$s = \begin{cases} \alpha r & ; 0 \leq r < a \\ \beta(r-a) + s_a & ; a \leq r < b \\ \gamma(r-b) + s_b & ; b \leq r \leq L-1 \end{cases}$$

where; α, β & γ are the slopes.

→ In the figure, 'r' represents the horizontal axis i.e. input gray level and 's' represents the vertical axis i.e. output gray level.

→ The darker gray level is obtained by making a slope < 1 and brighter ~~st~~ obtained by making a slope > 1 .

2.1.3. Thresholding (Binary Image):



(Fig. Thresholding)

- It is the simplest method of the image processing that can be used to create binary image.
- If we observe the contrast stretching diagram, we notice that the first and last slopes are made zero.
- If $r_1 = r_2$ & $s_1 = 0$, $s_2 = L-1$, then the threshold function is given by

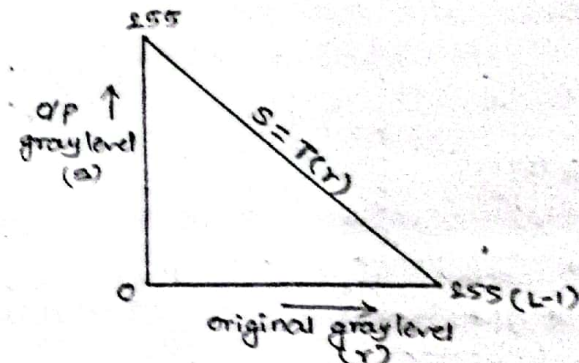
$$\begin{aligned} s &= 0; r < b \Rightarrow 0 \\ s &= L-1; r \geq b \Rightarrow 1 \end{aligned} \quad \text{where, } \begin{aligned} b &= \text{thresholding value} \\ L &= \text{max. gray level} \end{aligned}$$

Note:

Threshold image has maximum contrast on it having black & white gray level only.

$$T(r) = \{(L-1) - r\}$$

2.1.4. Digital Negative (X-Ray):



(Fig. Digital Negative)

- Digital Negative is useful in a lots of application. Common example of digital negative is X-Ray image.
- As the name suggest, negative means simply inverting the gray level i.e. black into white and vice-versa.
- So the negative of digital image with gray level in the range from 0 to (L-1) is obtained by using negative transformation.

$$\text{i.e. } s = T(r) = \{(L-1) - r\}$$

where, r = input pixel value

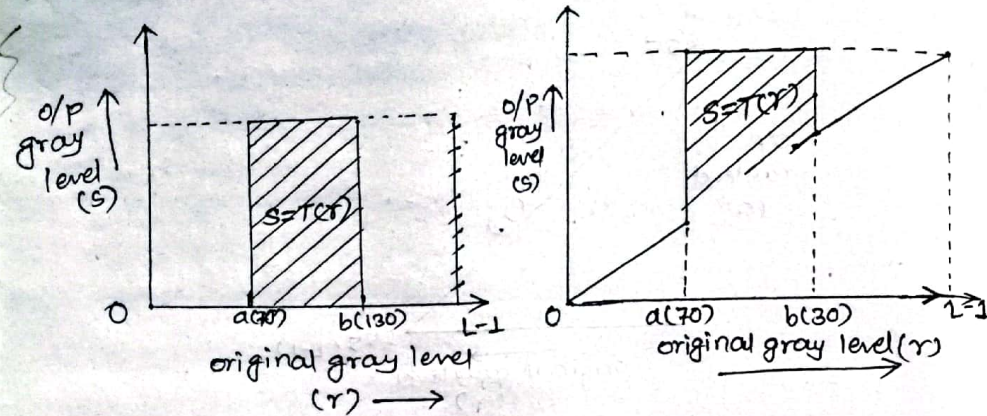
L = No. of gray level / Max. gray level

$T(r)$ = Transformation function

s = output gray level

- when $r=0$, $s=255$ i.e. white
- when $r=255$, $s=0$ i.e. black

2.1.5. Intensity Level Slicing (Gray Level Slicing): v. Imp



Fig(i). Slicing without background

Fig(ii). Slicing with background

→ Intensity level slicing is also known as gray level slicing that uses transformation function on a gray level.

→ There are two types of slicing:-

i> Slicing without background;

→ In this slicing, we have lost completely the background as shown in fig. (i) during enhancement.

→ This can be implemented by using the following function,

$$\text{i.e. } s = L-1 ; a \leq r \leq b \\ s = 0 ; \text{otherwise}$$

ii> Slicing with Background:

→ In this slicing, we haven't lost the background as shown in fig (ii).

→ In this, we need not only to enhance the band of gray level but also need to retain the background.

→ This can be implemented by following function,

$$\text{i.e. } s = L-1 ; a \leq r \leq b$$

$$s = r ; \text{otherwise}$$

2.1.6. Bit Plane Slicing (Clipping):

→ Bit plane slicing is the clipping method that is used to enhance the features with an image.

→ We must provide a threshold level to determine how the clipping occurs.

→ The threshold level or value is a reference value where other values get changed.

→ In general, it is performed for maximum and minimum operators and expression that contain an image with threshold levels.

→ For example;

If we have image variable and want to slice it to ≥ 125 , then the resulting image stated as:

$$\text{clipped image} = [(\text{Image}(\text{Original}) > 125)]$$

It means if the element values is less than 125, then it is set as equal to 125; otherwise it is unchanged.

2.2. Introduction to Histogram:

- Histogram means global description of the appearance of the image.
- The information obtained from histogram is innermost and hence, histogram modeling is important in digital image processing.
- By the definition of histogram of an image, represents the relative frequency of occurrence of various gray level in an image.
- Histogram of an image can be plotted as:
 - i) The x-axis has the gray level and y-axis has the number of pixels in each gray level.
 - ii) The x-axis represent gray level and y-axis represent the probability of occurrence of that gray level.

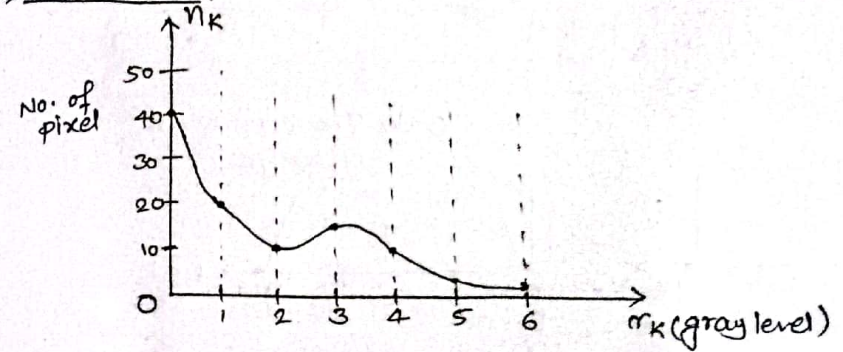
→ The method of histogram is shown below:

step 1: Original image:

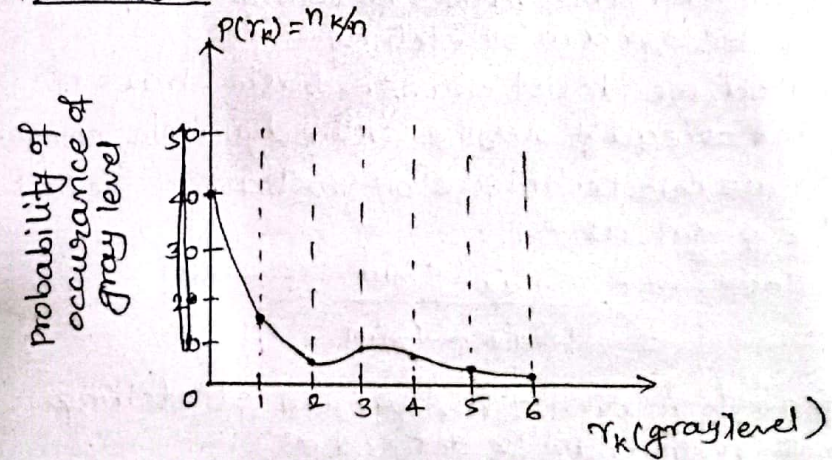
Gray level (r_k)	No. of Pixel (n_k)
0	40
1	20
2	10
3	15
4	10
5	3
6	2

step 2: Histogram Modeling:

i) Method 1:



ii) Method 2:



where,

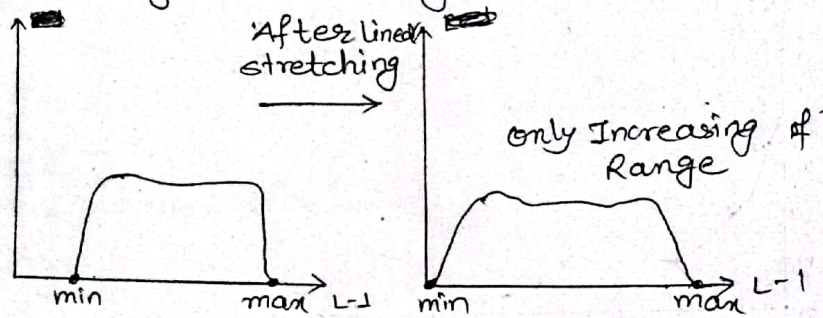
$r_k = k^{\text{th}}$ gray level

$n_k = \text{no. of pixel in } k^{\text{th}} \text{ gray level}$

$n = \text{total no. of pixel.}$

2.3. Histogram Techniques:

2.3.1. Histogram Stretching / Linear stretching:



(Fig. Linear stretching)

- To increase the dynamic range by using a technique is known as histogram stretching.
- In this method, we do not change basic shape of histogram but spread it over the entire dynamic range.
- For this, we use this technique on the basis of slope.
- It can be given as:-

$$\text{slope, } m = \frac{S_{\max} - S_{\min}}{r_{\max} - r_{\min}} \quad \text{--- (i)}$$

- For the transformation of histogram stretching, the transformation can be defined as:

$$s = T(r) = \frac{S_{\max} - S_{\min}}{r_{\max} - r_{\min}} (r - r_{\min}) + S_{\min}$$

$$\therefore s = \frac{S_{\max} - S_{\min}}{r_{\max} - r_{\min}} (r - r_{\min}) + S_{\min}$$

where,

$S_{\max}, S_{\min}$ = maximum & minimum gray level in o/p image

$r_{\max}, r_{\min}$ = maximum & minimum gray level in I/P image

- The transformation function stretches or increases the dynamic range of given image.

Q: Perform linear stretching of the following image on the basis of dynamic range i.e. $[0, 7]$.

Gray level, r_k	0	1	2	3	4	5	6	7
No. of pixel, n_k	0	0	50	60	50	20	10	0

Here,

$$r_{\min} = 2 ; r_{\max} = 6$$

$$S_{\min} = 0 ; S_{\max} = 7$$

We know,

$$s = T(r) = \frac{S_{\max} - S_{\min}}{r_{\max} - r_{\min}} (r - r_{\min}) + S_{\min}$$

$$= \frac{7-0}{6-2} (r-2) + 0$$

$$\therefore s = \frac{7}{4} (r-2)$$

Since the range of r is $(2, 6)$.

$$\text{when } r=2; s = \frac{7}{4} (2-2) = 0$$

$$\text{when } r=3; s = \frac{7}{4} (3-2) = \frac{7}{4} \approx 1.75 \approx 2$$

$$\text{when } r=4; s = \frac{7}{4} (4-2) = 3.5 \approx 4$$

$$\text{when } r=5; s = \frac{7}{4} (5-2) = \frac{21}{4} \approx 5.25 \approx 5$$

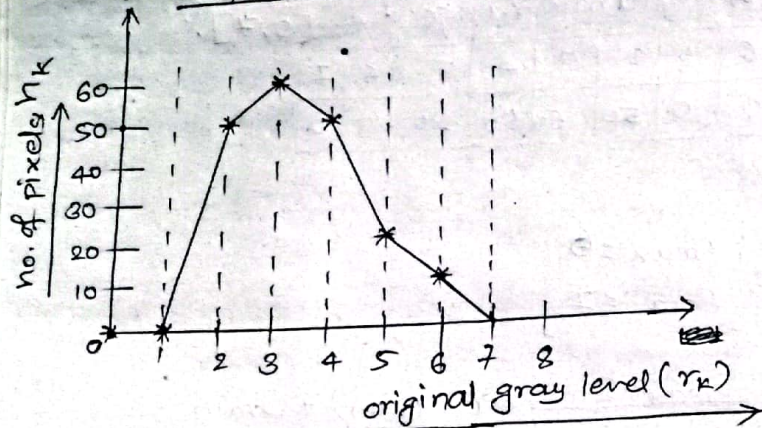
$$\text{when } r=6; s = \frac{7}{4} (6-2) = \frac{7}{4} \times 4 \approx 7$$

Hence, the modified gray level histogram or stretched histogram can be illustrated as:

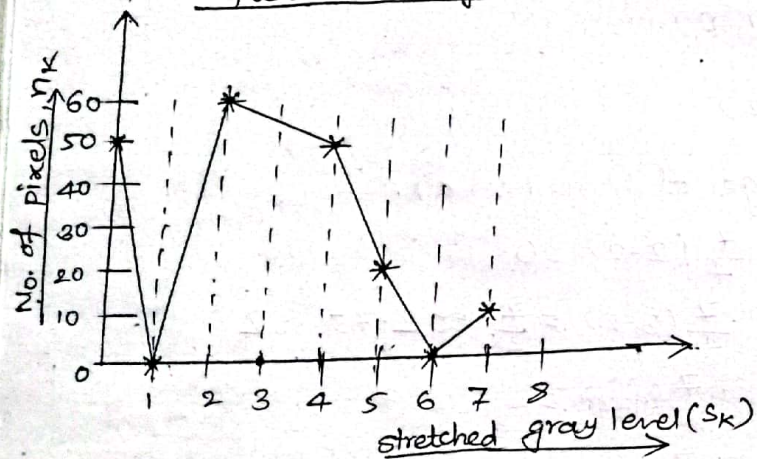
Gray level, s_k	0	1	2	3	4	5	6	7
No. of pixel, n_k	50	0	60	0	50	20	0	10

2.3.1

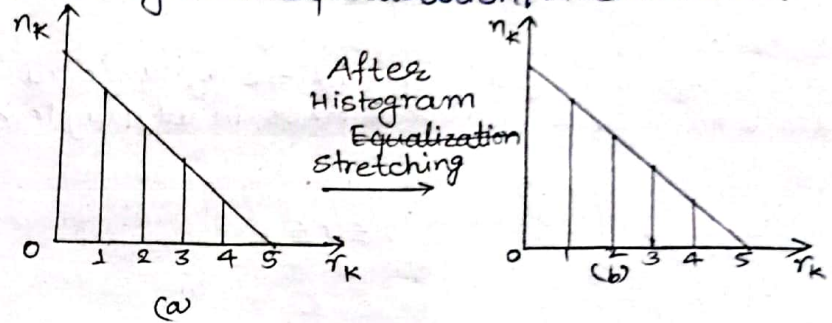
Before stretching



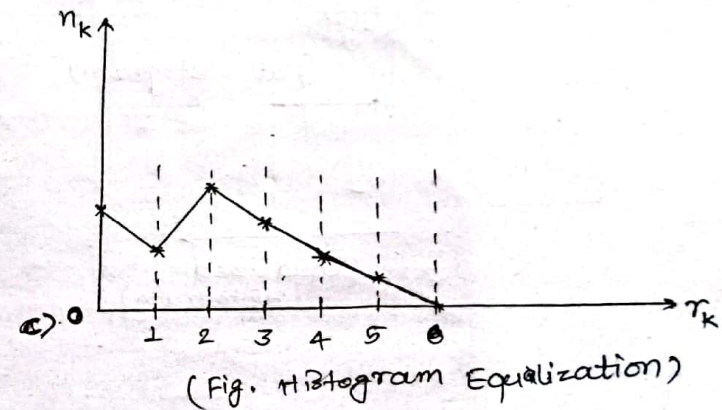
After stretching



2.3.2. Histogram Equalization/Linearization:



After Histogram Equalization, not increase dynamic range but spread uniform pixel in image



- The gray level for continuous variable can be characterized by probability density function (pdf) i.e. $P_r(r)$ & $P_s(s)$.
- We need to find the transformation which could be the flat histogram.
- Linear stretching is a good technique but shape remain same.
- There are many application where need a flat histogram so that this cannot be achieved by histogram stretching.
- Hence, a technique can be used to obtain a uniform histogram is known as histogram equalization or histogram linearization.

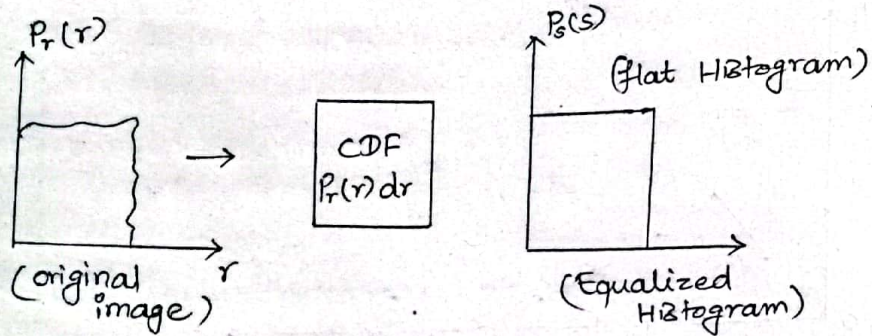
→ A perfect image is one which equal number of pixels in all gray level.

→ Therefore, image is not only spread the dynamic range but also to have equal number of pixels in all gray level.

We know that,

$$S = T(r) = \int_0^r P_r(r) ; 0 \leq r \leq 1$$

→ Above equation shows the transformation function of cumulative pdf (cdf) and it can be shown as:



1. Equalize the given histogram.

Gray level, r_k	0	1	2	3	4	5	6	7
No. of pixel, n_k	790	1023	850	656	329	245	122	81

Step 1:

Gray level, r_k	n_k	P.D.F = $P_r(r_k) = n_k/n$	C.D.F = $S_k = \sum P_r(r_k)$	$7 * S_k$	Rounding off
0	790	0.19	0.19	1.33	1
1	1023	0.25	0.44	3.08	3
2	850	0.21	0.65	4.55	5
3	656	0.16	0.81	5.67	6
4	329	0.08	0.89	6.23	6
5	245	0.06	0.95	6.65	7
6	122	0.03	0.98	6.86	7
7	81	0.02	1.00	7	7
	$n = 4096$				

Step 2: Modified Histogram:

Gray level, s_k	0	1	2	3	4	5	6	7
No. of pixel, n_k	0	790	0	1023	0	850	985	448

Hence, the equalized histogram:

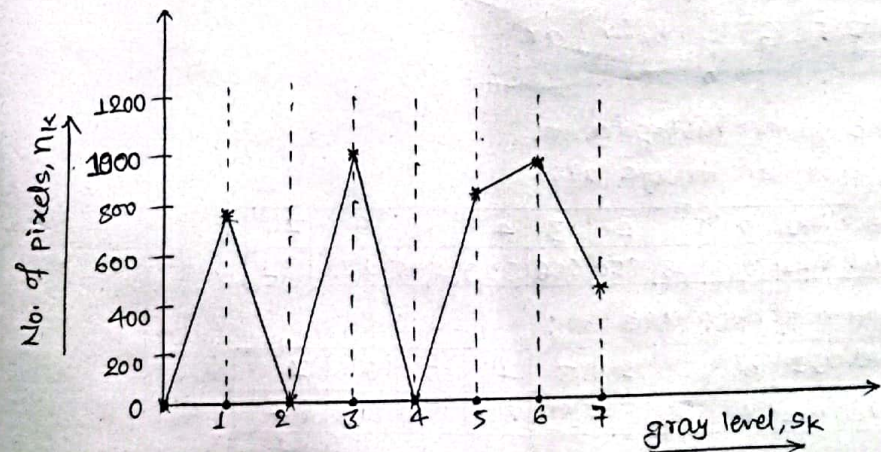


Fig. Equalized Image/Histogram

2-3-3. Histogram Specification/Histogram Matching:

→ Histogram matching is not interactive because it always gives one result i.e. An approximation to an uniform histogram.

→ At time, it is desirable to have an interactive methods in which certain gray level are highlighted.

→ Let us suppose, $P_r(r)$ is the original p.d.f,
 $P_z(z)$ is the desired p.d.f.

→ Suppose that histogram equalization is the first applied on the original image i.e.

$$s = T(r) = \int_0^r P_r(r) dr$$

→ Histogram equalization of desired image 'z' is that,
 i.e. $v = G(z) = \int_0^z P_z(r) dz$

→ Now, the inverse process $z = G^{-1}(s)$ which have the desired p.d.f. that provide histogram specification,

$$\text{i.e. } z = G^{-1}(s) = G^{-1}(T(r))$$

$$z = G^{-1} \int_0^r P_r(r) dr$$

→ Match the given histograms:

i> Histogram for image (a);

Gray level, r_{k1}	0	1	2	3	4	5	6	7
No. of pixel, n_{k1}	790	1023	850	656	329	245	122	81

ii> Histogram for image (b);

Gray level, r_{k2}	0	1	2	3	4	5	6	7
No. of pixel, n_{k2}	0	0	0	614	819	1230	819	614

Step 1: Histogram equalization of image (a) & (b).

i> Equalize histogram of image (a):

Gray level, r_{k1}	No. of Pixel, n_{k1}	$P_r(r_{k1}) = \frac{n_{k1}}{n}$	$S_{k1} = \sum P_r(r_{k1})$	$7 \times S_{k1}$	Rounding off
0	790	0.19	0.19	1.33	1 ✓
1	1023	0.25	0.44	3.08	3 ✓
2	850	0.21	0.65	4.55	5 ✓
3	656	0.16	0.81	5.67	6 } ✓
4	329	0.08	0.89	6.23	6 } ✓
5	245	0.06	0.95	6.65	7 } ✓
6	122	0.03	0.98	6.86	7 } ✓
7	81	0.02	1.00	7	7 } ✓
	$n = 4096$				

Now,

Gray level, s_{k1}	0	1	2	3	4	5	6	7
No. of pixel, n_{k1}	0	790	0	1023	0	850	985	448

ii> Equalize histogram of image (b):

Gray level, r_{k2}	No. of Pixel, n_{k2}	$P_z(r_{k2}) = \frac{n_{k2}}{n}$	$S_{k2} = \sum P_z(r_{k2})$	$7 \times S_{k2}$	Rounding off
0	0	0	0	0	0
1	0	0	0	0	0
2	0	0	0	0	0
3	614	0.14	0.14	0.98	1
4	819	0.19	0.33	2.31	2
5	1230	0.3	0.63	4.41	4
6	819	0.19	0.82	5.74	6
7	614	0.14	0.96	6.72	7
	$n = 4096$				

Now 1

Gray level, s_k	0	1	2	3	4	5	6	7
No. of pixel, n_k	0	614	819	0	1230	0	819	614

Step 2: To obtain histogram specification or matching, we apply inverse transformation comparing both equalized histogram.

Gray level, s_k	0	1	2	3	4	5	6	7
No. of pixel, n_k	0	0	0	1023	0			

2.4. Spatial Operations:

→ Many image enhancement techniques are based on spatial operation.

→ Most of the spatial operation are performed on the basis of local neighborhood of the input pixel.

→ Some of the example of spatial operation for image enhancement are as follows:-

i) Neighborhood Processing

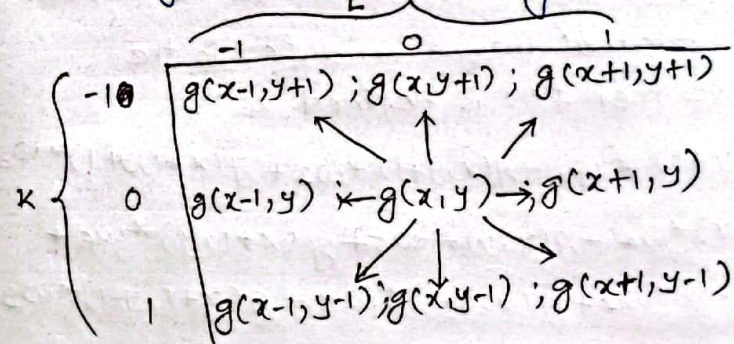
ii) Spatial averaging

iii) Smoothing spatial filters

iv) Zooming

v) Low pass, High Pass & Band Pass filters.

2.4.1. Neighborhood Processing:



(Fig. 3*3 neighborhood Pixel)

→ We change the value of pixel based on the values of its eight (8) neighbors as shown above.

→ Instead of 3*3 neighborhood, we can also use 5*5, 7*7, etc.

→ Neighborhood processing deals with the point operation.

→ The neighborhood pixel shown above can be masked into window as shown in figure below:

		L		
		-1	0	1
K	-1	w_1	w_2	w_3
	0	w_4	w_5	w_6
	1	w_7	w_8	w_9

(Fig. 3*3 window)

→ The above figure is a type of template which is also called as mask or window.

→ To achieve neighborhood processing, we place this 3*3 mask on the image corresponding to each of the component and get some information at the centre.

→ If $g()$ is the original image and $f()$ is the modified image then $f()$ is defined as;

$$f(x, y) = g(x-1, y+1) * w_1 + g(x, y+1) * w_2 + g(x+1, y+1) * w_3 + g(x-1, y) * w_4 + g(x, y) * w_5 + g(x+1, y) * w_6 + g(x-1, y-1) * w_7 + g(x, y-1) * w_8 + g(x+1, y-1) * w_9$$

→ Hence, the sum of the product of the mask coefficients with corresponding pixel take place.

2.4.2. Spatial Averaging / Averaging Filtering:

→ In spatial averaging, each pixel is replaced by the weighted average of its neighborhood pixel,

$$i.e. v(m, n) = \sum_{(K, L) \in W} a(K, L) y(m-K, n-L)$$

where,

$y(m-K, n-L)$ = input image

$v(m, n)$ = output image

W = chosen window

$a(K, L)$ = filter weight

→ A common class of spatial average filter has all equal weight,

$$i.e. v(m, n) = \frac{1}{N_w} \sum_{(K, L) \in W} y(m-K, n-L)$$

where,

$$a(K, L) = \frac{1}{N_w}, \quad N_w = \text{no. of pixel in the window}$$

→ Now, each pixel is replaced by its average with the average of its nearest pixel value.

→ Spatial averaging is used for noise smoothing, low pass filtering and sub sampling of the image.

→ Example:

		L=3		
		-1	0	1
K=3	-1	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
	0	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
	1	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

(Fig. 3*3 mask window)

		L=2	
		-1	1
K=2	-1	$\frac{1}{4}$	$\frac{1}{4}$
	1	$\frac{1}{4}$	$\frac{1}{4}$

(Fig. 2*2 mask window)

2.4.3. Smoothing Spatial Filter:

2.4.3.1. Median Filter:

- The median filter is also known as a spatial filter but it replaces the central value in the window with median of all the pixel value in the window.
- The kernel or template or mask is usually square but can be initiated in different window.
- For example;

6	2	0
3	97	4
19	15	10

i.e. 6, 2, 0, 3, 97, 4, 19, 15, 10

Arranging the above numbers in ascending order,

0, 2, 3, 4, 6, 10, 15, 19, 97

$$\therefore M_d = \text{o/p value} = \left(\frac{n+1}{2}\right)^{\text{th item}} = \left(\frac{9+1}{2}\right)^{\text{th item}} = 5^{\text{th item}}$$

i.e. $M_d = 6$

After filtering, the median central value is replaced by 6 i.e. $97 \Rightarrow 6$.

6	2	0
3	6	4
19	15	10

→ Formula for median filtering is;

$$v(m, n) = \text{Median} \{ y(m-k, n-l) \mid (k, l) \in w \}$$

2.4.3.1. Mean Filter:

- Mean filter is a spatial filter that replaces the central value in the window with the average of all the pixel values in the window.
- The window can be of any shape but can be initiated as a square mask.
- Therefore, the average of smoothing filter shows the properties of low pass filter. So, it reduces the noise from an image.
- For example;

5	3	6
2	1	9
8	4	7

i.e. 5, 3, 6, 2, 1, 9, 8, 4, 7

$$\therefore \text{Mean, } \bar{x} = \text{o/p value} = \frac{\sum x}{N} = \frac{45}{9} = 5$$

- After filtering, the central value of the given window ~~can~~ be replaced by the mean value of the square i.e. 5.

6	2	0
2	5	9
8	4	7

2.4.4. Zooming!

→ Zooming is another important application in image enhancement where spatial domain neighborhood operation is used.

→ Actually, zooming is the process of making an image larger.

→ Zooming involves following two steps:

i) Creating of new pixel location

ii) Assigning of gray level to those new location.

→ The different types of zooming techniques are as:

i) Replication

ii) Linear Interpolation.

2.4.4.1. Replication:

→ In replication, we simply replicate each pixel and then replicate row.

→ Let us consider an image as:

1	2	3	4
5	6	7	8
9	8	6	7
0	1	2	3

→ As start from the first row, first we replicate each pixel and then replicate each row.

→ The first row looks like

$[1 \ 1 \ 2 \ 2 \ 3 \ 3 \ 4 \ 4]$

→ We will now replicate this row then it looks as:

$\begin{bmatrix} 1 & 1 & 2 & 2 & 3 & 3 & 4 & 4 \\ 1 & 1 & 2 & 2 & 3 & 3 & 4 & 4 \end{bmatrix}$

→ Now, performing replication operation on an entire image then we obtain output image as

1	1	2	2	3	3	4	4
1	1	2	2	3	3	4	4
5	5	6	6	7	7	8	8
5	5	6	6	7	7	8	8
9	9	8	8	6	6	7	7
9	9	8	8	6	6	7	7
0	0	1	1	2	2	3	3
0	0	1	1	2	2	3	3

→ Here, 4×4 image is zoomed to 8×8 image.

→ This method can be repeated to get larger image but no new data is added.

Zooming by zero Interlace Technique

Step 1: Adding zero to every after pixel of the first row, then we obtain row as:

$[1 \ 0 \ 2 \ 0 \ 3 \ 0 \ 4 \ 0]$

Step 2: Adding zero at every column, i.e. insert row of zero & then this is also known as zero-interlacing.

$\begin{bmatrix} 1 & 0 & 2 & 0 & 3 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Step 3: Performing zero-interlace technique on an entire image then we obtain output image.

1	0	2	0	3	0	4	0
0	0	0	0	0	0	0	0
5	0	6	0	7	0	8	0
0	0	0	0	0	0	0	0
9	0	8	0	6	0	7	0
0	0	0	0	0	0	0	0
0	0	1	0	2	0	3	0
0	0	0	0	0	0	0	0

(Fig. 8x8 final pseudo image)

→ Hence, this image is known as zero-interlace image.

2.4.4.2. Linear Interpolation:

→ In linear interpolation, instead of replicating each pixel, average of two adjacent pixel along the row is taken and placed between two pixels.

→ Accordingly same operation is then performed along the column.

→ Linear interpolation is also derived from the zero-interlace technique.

Step 1: Zero Interlace:

1	0	2	0	3	0	4	0
0	0	0	0	0	0	0	0
5	0	6	0	7	0	8	0
0	0	0	0	0	0	0	0
9	0	8	0	6	0	7	0
0	0	0	0	0	0	0	0
0	0	1	0	2	0	3	0
0	0	0	0	0	0	0	0

Step 2: Interpolate row;

1	1.5	2	2.5	3	3.5	4	2
0	0	0	0	0	0	0	0
5	5.5	6	6.5	7	7.5	8	4
0	0	0	0	0	0	0	0
9	8.5	8	7	6	6.5	7	3.5
0	0	0	0	0	0	0	0
0	0.5	1	1.5	2	2.5	3	1.5
0	0	0	0	0	0	0	0

Step 3: Interpolate Column:

1	1.5	2	2.5	3	3.5	4	2
3	3.5	4	4.5	5	5.5	6	3
5	5.5	6	6.5	7	7.5	8	4
7	7	7	6.25	6.5	7	7.5	3.25
9	8.5	8	7	6	6.5	7	3.5
4.5	4.5	4.25	4.5	4	4.5	5	2.5
0	0.5	1	1.5	2	2.5	3	1.5
0	0.25	0.5	0.75	1	1.25	1.5	0.75

2.4.5. Filtering:

2.4.5.1. Low Pass Filtering:

- Low pass filtering is employed to remove high frequency (unnecessary noise) from a digital image.
- This type of filtering is usually used to attenuate the image noise.
- Here, an image is smoothened by decreasing the disparity between the pixel value by average value.
- Example of mask or kernel or window of low pass filter can be shown as below:

$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

(fig. 3*3 mask)

$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$

(fig. 4*4 mask)

2.4.5.2. High Pass Filtering:

- A pass filtering is the basis for most sharpening methods.
- Sharpening is the opposite of smoothing.
- The sharpening can make it possible to highlight the borders between homogeneous region.
- This is used to return high frequency information.
- The kernel of the high pass filter is designed to increase the brightness of center pixel.
- Example of mask or kernel or window of high pass filter is as shown below:

$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

(fig. 3*3 mask high pass filter)

2.4.5.3. Band Pass Filtering:

- Band pass filtering is used to remove selected frequency region between low and high frequency.
- It is used for image restoration.
- It passes a limited range of frequency.
- Calculating spatial averaging for band pass filtering is;

$$Y(m, n; 0) = \frac{1}{N_0} \sum_{(k, l) \in W_0} Y(m-k, n-l)$$

where,

W_0 = Neighborhood mask along directions

N_0 = No. of pixel within the neighborhood

3. Image Enhancement in the Frequency Domain (6 hrs)

3.1. Introduction:

- In the frequency domain, a digital image is converted from spatial domain to frequency domain and image filtering is used for image enhancement for a specific application.
- There are two types of filters. They are low pass & high pass filters.
- Low pass filter is used for smoothening an image.
- High pass filter is used for sharpening an image.
- The low pass filters and low pass filters have sub-categorized i.e. ideal filter, Butterworth filter and Gaussian filter.

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- There are two types of filters. They are low pass & high pass filters.
- Low pass filter is used for smoothening an image.
- High pass filter is used for sharpening an image.
- The low pass filters and high pass filters have sub-categorized i.e. ideal filter, Butterworth filter and Gaussian filter.
- The ideal lowpass filter is used to cut-off all the high-frequency components of Fourier Transform.
- Butterworth Lowpass filter is used to remove high-frequency noise with very minimal loss of signal components if ~~order~~ is high; otherwise it reduces the noise and smoothen the transition if order is low.
- A Gaussian lowpass filter is used for smoothen the transition and blurring regions of an image.
- The ideal highpass filter is used to cut-off all the low frequency components of Fourier Transform.
- Butterworth highpass filter is used to remove low-frequency noise with very minimal loss of signal components if order is low; otherwise it reduces the noise and sharpens the edges if ~~the~~ order is high.
- A Gaussian high pass filter is used for sharpening the edges.

8.2. Image Enhancement in the frequency domain

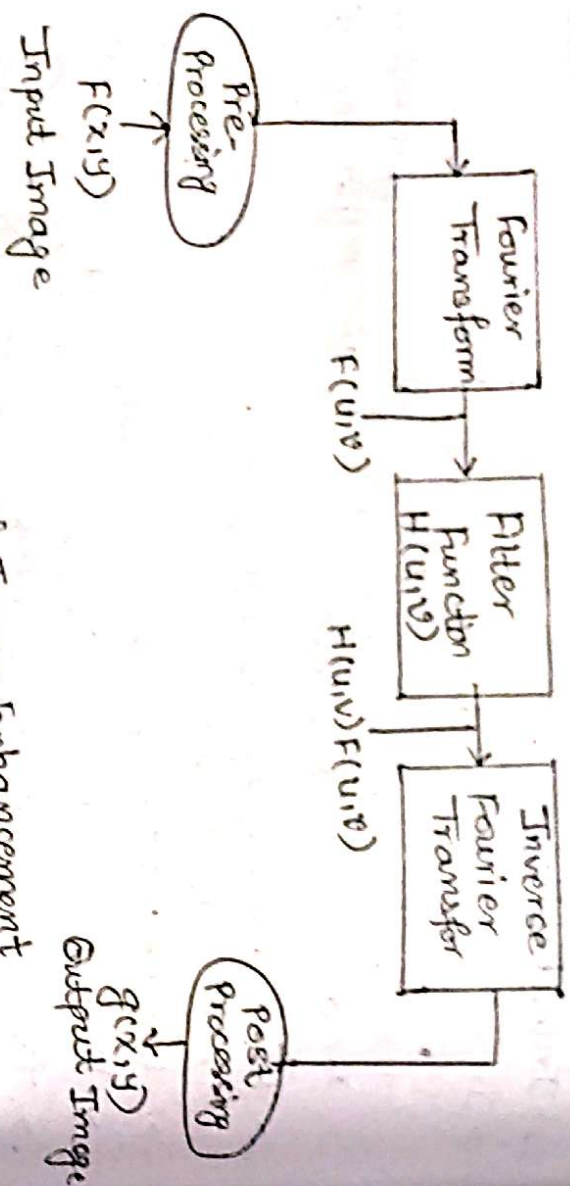


Fig. steps of Image Enhancement in frequency domain

i) Pre-Processing:

- Pre-processing is applied to the image to prepare it for filtering by addressing any issue or enhancing certain features.
- In this step, gray scale conversion, noise reduction and contrast enhancement are carried out.

ii) Fourier Transform:

- Fourier Transform is applied to convert the image from spatial domain to frequency domain.
- This helps to analyze and manipulate the frequency content of the image.

i.e. $F(u,v) = \text{Fourier Transform [pre-processed Image } (x,y)]$

iii) Filter function:

- Once the image is in frequency domain, a filter function is applied to modify the frequency components.

i.e. $G(u,v) = F(u,v) \times \text{Filter Function of } H(u,v)$

iv) Inverse Fourier Transform:

- The inverse Fourier transform converts the filtered image back to the spatial domain.

i.e. $\text{Processed Image } (x,y) = \text{Inverse Fourier Transform of } [G(u,v)]$

v) Post-Processing:

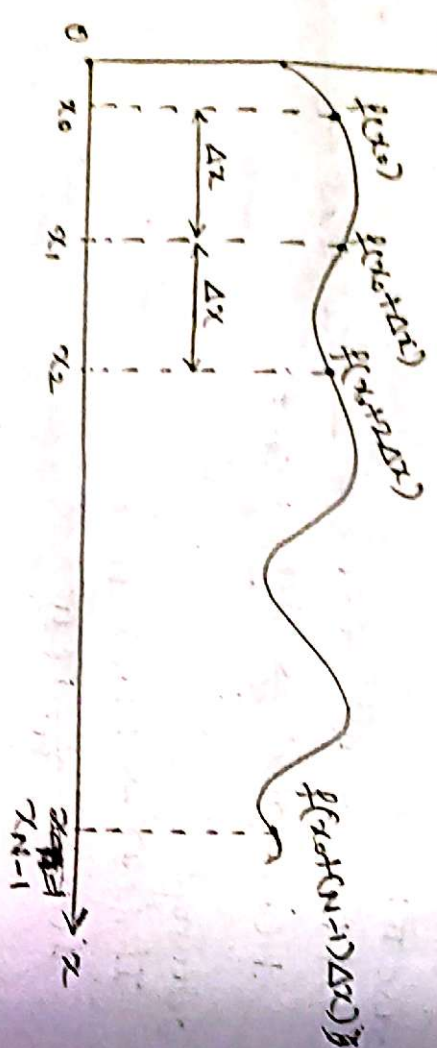
- Post processing is applied for final adjustment of the image.
- In this step, Normalizing, Adjusting Contrast & Brightness, Additional filtering are done.

i.e. $\text{Final Image } g(x,y) = \text{Post Process [Processed Image } (x,y)]$

3.4. Discrete Fourier Transform (DFT):

→ Suppose a continuous function, $f(x)$ is discretized in a sequence,

$\{f(x_0), f(x_0 + \Delta x), f(x_0 + 2\Delta x), \dots, f(x_0 + (N-1)\Delta x)\}$ by taking N samples Δx unit apart as shown in figure below.



→ The DFT is defined by following way:-

i) For 1-D DFT:

$$F(n) = \sum_{k=0}^{N-1} f(k) e^{-j2\pi \frac{nk}{N}} ; n = 0, 1, 2, \dots, N-1.$$

ii) For 2-D DFT:

$$F(m, n) = \sum_{k=0}^{M-1} \sum_{l=0}^{N-1} f(k, l) e^{-j2\pi \left(\frac{mk}{M} + \frac{nl}{N} \right)}$$

→ The DFT and Inverse DFT is given as:

i) For 1-D:

$$F(u) = \sum_{x=0}^{N-1} f(x) e^{-j2\pi \frac{ux}{N}} ; u = 0, 1, \dots, N-1$$

$$f(x) = \frac{1}{N} \sum_{u=0}^{N-1} F(u) e^{j2\pi \frac{ux}{N}} ; x = 0, 1, \dots, N-1$$

ii) For 2-D:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

for $u = 0, 1, 2, \dots, M-1$ & $v = 0, 1, 2, \dots, N-1$

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

for $x = 0, 1, 2, \dots, M-1$ & $y = 0, 1, 2, \dots, N-1$

→ The Fourier spectrum is given by

$$F(u, v) = \sqrt{R^2(u, v) + I^2(u, v)} ; F(u) = \sqrt{R^2(u) + I^2(u)}$$

→ The Phase Angle is given by

$$\phi(u) = \tan^{-1} \left(\frac{I(u)}{R(u)} \right) \text{ & } \phi(u, v) = \tan^{-1} \left(\frac{I(u, v)}{R(u, v)} \right)$$

→ The power spectrum is given by

$$|P(u)| = |F(u)|^2 = R^2(u) + I^2(u)$$

$$|P(u, v)| = |F(u, v)|^2 = R^2(u, v) + I^2(u, v)$$

→ The transform matrix

$$M = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -j \\ 1 & j & -1 & -j \end{bmatrix} = M^T$$

4x4

To calculate the DFT using matrix method,

i) For 1D Image;

$$F = M \cdot f ; f = \text{given matrix}$$

ii) For 2D Image;

$$F = M F M^T ; f = \text{given matrix}$$

* Properties of DFT:

i) Linearity:

⇒ If $f_1(x) \rightarrow F_1(\omega)$ and $f_2(x) \rightarrow F_2(\omega)$, then

$$a f_1(x) + b f_2(x) \rightarrow a F_1(\omega) + b F_2(\omega)$$

where a & b are constants.

ii) Periodicity:

⇒ If $f(x+N) \rightarrow f(x)$ for all x then

$$F(x+N) \rightarrow F(x) \text{ for all } x$$

iii) Circular Time Shift:

⇒ If $f(x) \rightarrow F(x)$ then

$$F(x) e^{-j2\pi \frac{ux}{N}} \rightarrow f(x-k)_{\text{all } x}$$

i) Conjugation:

⇒ If $f(x) \rightarrow F(x)$ then,

$$f^*(x) \rightarrow F^*(N-x)$$

ii) Circular Frequency Shift:

⇒ If $f(x) \rightarrow F(x)$ then,

$$f(x) \cdot e^{j2\pi \frac{ux}{N}} \rightarrow F[(x-k)]_N$$

iii) Parseval's Theorem:

⇒ If $f_1(x) \rightarrow F_1(x)$ & $f_2(x) \rightarrow F_2(x)$ then by Parseval's Theorem,

$$\sum_{x=0}^{N-1} f_1(x) f_2^*(x) = \frac{1}{N} \sum_{u=0}^{N-1} F_1(u) F_2^*(u)$$

* Applications of DFT:

i) Signal Analysis

ii) Sound Filtering

iii) Data Compression

iv) Cross correlation

v) Matched Filtering

vi) System Identification

vii) Power spectrum estimation

viii) Display signal and spectrum.

3.5. Smoothing frequency domain

- Smoothing is achieved by in the frequency domain by dropping out the high frequency components.
- The basic model for filtering is given by

$$G(u,v) = H(u,v)F(u,v)$$

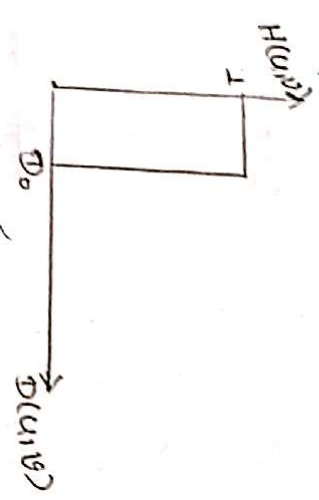
where, $F(u,v)$ = Fourier Transform of the image being filtered

$H(u,v)$ = Filter Transform Function.

$$H(u,v) = \text{Filter Transform Function.}$$

- Low pass filters only pass the low frequencies, drop the high ones.

- Ideal low pass filter is used to cut all high frequency components that are a specified distance D_0 from the origin of transform.



- Transfer function;

$$H(u,v) = \begin{cases} 1 & ; D(u,v) \leq D_0 \\ 0 & ; D(u,v) > D_0 \end{cases}$$

where $D(u,v)$ is given as

$$D(u,v) = \left[\left(u - \frac{M}{2} \right)^2 + \left(v - \frac{N}{2} \right)^2 \right]^{1/2}$$

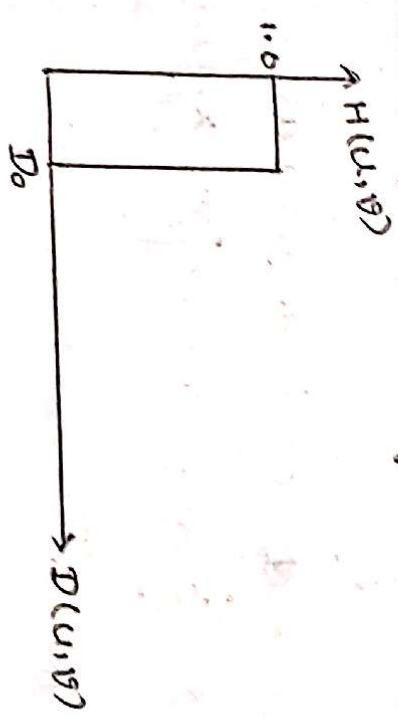
3.6. Sharpening frequency domain Filter: (HPF).

- Sharpening is achieved in the frequency domain by dropping out the low pass frequency components.
- High pass filters only pass the high frequencies, drop the low ones.

- The high pass frequencies are precisely the reverse of low pass filters, so,

$$H_{hp}(u,v) = 1 - H_{lp}(u,v)$$

- The ideal high pass filter is used to cut low frequency components that are a specified distance D_0 from the origin of transform.



- Transfer function;

$$H(u,v) = \begin{cases} 0 & ; D(u,v) \leq D_0 \\ 1 & ; D(u,v) > D_0 \end{cases}$$

3.6. Other Image Transformations

3.6.1. Discrete Cosine Transform (DCT):

→ Discrete cosine transform converts the sequence of values into set of cosine function.

→ DCT is only concerned with real numbers.

→ DCT is defined as,

$$DCT = C(u) = \alpha(u) \sum_{x=0}^{N-1} f(x) \cos \left[\frac{\pi(2x+1)u}{2N} \right]$$

where, $u = 0, 1, 2, \dots, N-1$

$$\alpha(u) = \sqrt{\frac{1}{N}} = \frac{1}{\sqrt{N}} \quad ; \quad u = 0$$

$$\alpha(u) = \frac{2}{\sqrt{N}} \quad ; \quad u = 1, 2, \dots, N-1$$

→ The Inverse DCT is defined as,

$$IDCT = f(x) = \sum_{u=0}^{N-1} \alpha(u) c(u) \cos \left[\frac{\pi(2x+1)u}{2N} \right]$$

where $u = 0, 1, 2, \dots, N-1$

ii) For 2-D:

$$DCT = c(u, v) = \alpha(u, v) \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \cos \left(\frac{\pi(2x+1)u}{2N} \right)$$

$$\cos \left(\frac{\pi(2x+1)u}{2N} \right)$$

where, $\alpha(u, v) = \frac{1}{N}$ for $u, v = 0$

$$= \sqrt{\frac{2}{N}} \quad \text{for } u, v = 1, \dots, N-1$$

$$u = v = 0, 1, 2, \dots, N-1$$

$$IDCT = f(x, y) = \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} \alpha(u, v) c(u, v) \cos \left(\frac{\pi(2x+1)u}{2N} \right)$$

$$\cos \left(\frac{\pi(2x+1)u}{2N} \right)$$

where, $u = v = 0, 1, 2, \dots, N-1$

* Properties of DCT:

i) The DCT is real and orthogonal i.e. $C = C^* \Rightarrow C^{-1} = C$

where, $C^* = \text{Conjugate of } C$

ii) The cosine transform is not real part of unitary DFT,

$$\text{i.e. } T^{-1} = T^* T$$

iii) DCT is a fast Transform. The cosine transform of 'N' elements can be calculated in $O(N \log_e N)$ operation through N points FFT.

iv) DCT has excellent Energy compaction for highly correlated data.

* Applications:

i) DCT is widely used in encoding, decoding, video, audio, multiplexing and analog to digital conversion.

ii) DCT is commonly used for HDTV encoder/decoder chips.

iii) DCT is used in JPEG compression as well.

→ The DCT Transform Matrix is given by

$$C = \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0.653 & 0.2305 & -0.2305 & -0.653 \\ 0.5 & -0.5 & -0.5 & 0.5 \\ 0.2305 & -0.653 & 0.653 & -0.2305 \end{bmatrix}_{4 \times 4}$$

To calculate the DCT using matrix method,

i) For 1D image:

$$F = C \cdot f \quad ; f = \text{given matrix}$$

For Inverse $f = F \cdot C$

ii) For 2D image

$$F = C \cdot f \cdot C^T \quad ; f = \text{given matrix}$$

$$\text{For Inverse } = C^T \cdot F \cdot C$$

3.6.2. Hadamard Transform:

→ Hadamard Transform is a mathematical operation that is used to convert a sequence of data into its frequency domain representation.

→ The Hadamard Transform is real, symmetric and orthogonal in nature.

→ Hadamard Transform has only binary values i.e. 1 or -1 in its kernel matrix.

→ The Hadamard Transform is defined as

i) For 1-D:

$$H(u) = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} f(x) (-1)^{\sum_{i=0}^{n-1} b_i(x) b_i(u)}$$

where, $N = 2^n$

$$u = 0, 1, 2, \dots, N-1$$

ii) For 2-D:

$$H(u, v) = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) (-1)^{\sum_{i=0}^{n-1} [b_i(x) b_i(u) + b_i(y) b_i(v)]}$$

where, $n = 2^n$

$$u, v = 0, 1, 2, \dots, N-1$$

→ The Inverse Hadamard Transform is given by

i) For 1-D:

$$f(x) = \sum_{u=0}^{N-1} H(u) (-1)^{\sum_{i=0}^{n-1} b_i(x) b_i(u)}$$

ii) For 2-D:

$$f(x, y) = \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} H(u, v) (-1)^{\sum_{i=0}^{n-1} [b_i(x) b_i(u) + b_i(y) b_i(v)]}$$

→ The Hadamard matrix

$$H_n = \frac{1}{\sqrt{N}} \begin{bmatrix} H_{n-1} & H_{n-1} \\ H_{n-1} & -H_{n-1} \end{bmatrix}$$

where $\frac{1}{\sqrt{N}}$ = Sample value

$$\begin{bmatrix} H_{n-1} & H_{n-1} \\ H_{n-1} & -H_{n-1} \end{bmatrix} = \text{Indicate Kernel}$$

i) For $n=1, N=2^1=2$:

$$H_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} H_{1-1} & H_{1-1} \\ H_{1-1} & -H_{1-1} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} H_0 & H_0 \\ H_0 & -H_0 \end{bmatrix}$$

$$\therefore H_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}_{2 \times 2}$$

ii) For $n=2, N=2^2=4$:

$$H_2 = \frac{1}{\sqrt{4}} \begin{bmatrix} H_{2-1} & H_{2-1} \\ H_{2-1} & -H_{2-1} \end{bmatrix} = \frac{1}{\sqrt{4}} \begin{bmatrix} H_1 & H_1 \\ H_1 & -H_1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{4}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}_{4 \times 4}$$

→ To calculate the Hadamard transform of the given image,

i) For 1D image:

$$g = H \times f \quad \& \quad \begin{bmatrix} f \\ f = H^T \times g \end{bmatrix} \quad (\text{for inverse})$$

ii) For 2-D image:

$$g = H \times f \times H^T \quad \& \quad \begin{bmatrix} f \\ f = H^T \times g \times H \end{bmatrix} \quad (\text{for inverse})$$

where, f = given matrix

g = calculated matrix

$$H = \frac{1}{\sqrt{4}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}_{4 \times 4} = H^T$$

* Properties of Hadamard Transform:

i) Hadamard Transform is real, symmetric and orthogonal in nature.

$$\text{i.e. } H = H^* \Rightarrow H^{-1} = H^T$$

ii) H is a fast transform and can be implemented in $O(N \log_2 N)$ addition & subtraction.

iii) H has only two binary values i.e. 1 or -1 in its kernel.

* Applications of Hadamard Transform:

i) H is used for digital image processing and digital signal processing.

ii) It preserves energy and had good energy compaction for highly correlated image.

iii) H is used in JPEG and MPEG compression as well.

3.6.3. Haar Transform: ~~Int~~

→ Haar Transform is the orthogonal transform which is derived from the Haar matrix.

→ It is also a fast transform and can be implemented in $O(N \log_2 N)$ operation.

→ It exists for $N = 2^n$ where n is an integer.

→ The Haar Transform is defined as:

$$H_T = H_K(x) ; K = 0, 1, 2, \dots, N-1$$

K is given as

$$K = 2^p + q - 1 \text{ and } 0 \leq p \leq (n-1)$$

$$1 \leq q \leq 2^p ; \text{ for } p \neq 0$$

$$q = 0, 1 ; \text{ for } p = 0$$

* Properties of Haar Transform:

i) It is symmetric, separable, unitary transform that uses Haar function for its basis.

ii) It is orthogonal and real i.e.

$$T^{-1} = T \text{ and } \Rightarrow H_T \Rightarrow H_T^*$$

iii) It is a fast transform and can be implemented in $O(N \log_2 N)$ operation where N is the no. of samples.

iv) It exists for $N = 2^n$ where n is an integer.

v) It has poor energy compaction property.

* Applications of Haar Transform:

i) JPEG & MPEG compression

ii) Television broadcasting

iii) Encoding and Decoding.

→ To calculate the Haar Transform of the given image,

i) For 1D-image:

$$g = H \times f$$

$$f = H^T \times g \text{ for inverse}$$

ii) For 2D-image:

$$g = H \times f \times H^T$$

$$f = H^T \times g \times H \text{ for inverse.}$$

where $f = \text{given image}$

$g = \text{obtain image}$

$$H = \frac{1}{\sqrt{4}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ \sqrt{2} & -\sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{2} & \sqrt{2} \end{bmatrix} 4 \times 4$$

$$H^T = \frac{1}{\sqrt{4}} \begin{bmatrix} 1 & 1 & \sqrt{2} & 0 \\ 1 & 1 & -\sqrt{2} & 0 \\ 1 & -1 & 0 & \sqrt{2} \\ 1 & -1 & 0 & -\sqrt{2} \end{bmatrix} 4 \times 4$$

For $N = 2 = 2^1$:

$$H_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} 2 \times 2$$

For $N = 4 = 2^2$:

$$H_2 = \frac{1}{\sqrt{4}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ \sqrt{2} & \sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{2} & \sqrt{2} \end{bmatrix} 4 \times 4$$

* Algorithm for Haar Transform: Kernels:

~~For N=2:~~

Step 1: Determine the order N of the Haar Basis.

Step 2: Determine n where $n = \log_2 N$.

Step 3: Determine p and q , where $0 \leq p \leq n-1$

i) If $p = 0$ then $q = 0$ or $q = 1$

ii) If $p \neq 0$ then $1 \leq q \leq 2^p$

Step 4: Determine K .

i.e. $K = 2^p + q - 1$

Step 5: Determine x .

$$x \rightarrow [0, 1] \rightarrow \left\{ \frac{0}{N}, \frac{1}{N}, \dots, \frac{N-1}{N} \right\}$$

Step 6:

$$h_{0,0}(x) = h_{0,1}(x) = \frac{1}{\sqrt{N}}, x \in [0, 1]$$

$$h_k(x) = h_{p,q}(x) = \frac{1}{\sqrt{N}} \begin{cases} 2^{p/2} & ; \frac{q-1}{2^p} \leq x \leq \frac{q-1/2}{2^p} \\ -2^{p/2} & ; \frac{q-1/2}{2^p} \leq x \leq \frac{q}{2^p} \\ 0 & ; \text{otherwise } x \in [0, 1] \end{cases}$$

The Haar Basis Matrix is given by

i) For $N=2$:

$$H_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

ii) For $N=4$:

$$H_4 = \frac{1}{\sqrt{4}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ \sqrt{2} & -\sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{2} & -\sqrt{2} \end{bmatrix}$$



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3.6. Fast Fourier Transform:

- FFT is an efficient algorithm for computing the discrete Fourier Transform (DFT) and its inverse.
- FFT reduces the computational complexity of DFT making it practical for real time signal processing.

→ The DFT is given by

$$F(u) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) e^{-j2\pi \frac{ux}{N}} ; u=0,1,\dots,N-1$$

$$\Rightarrow F(u) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) e^{j2\pi \frac{ux}{N}} \quad (\because W_N = e^{-j2\pi/N})$$

For some even no. of K , $N=2K$ then,

$$F(u) = \frac{1}{2K} \sum_{x=0}^{2K-1} f(x) (W_{2K})^{ux}$$

Separating K -even terms and K -odd terms, then,

$$F(u) = \frac{1}{2} \left\{ \frac{1}{K} \sum_{x=0}^{K-1} f(2x) (W_{2K})^{u(2x)} + \frac{1}{K} \sum_{x=0}^{K-1} f(2x+1) (W_{2K})^{u(2x+1)} \right\}$$

where,

$$W_{2K}^{2u} = W_{2K}^{2u} \quad \& \quad W_{2K}^{2(u+1)} = W_{2K}^{2u} W_{2K}^2 = W_{2K}^2 W_{2K}^{2u}$$

$$\therefore F(u) = \frac{1}{2} \left\{ \frac{1}{K} \sum_{x=0}^{K-1} f(2x) W_{2K}^{ux} + \frac{1}{K} \sum_{x=0}^{K-1} f(2x+1) W_{2K}^{ux} W_{2K}^u \right\}$$

$$\Rightarrow F(u) = \frac{1}{2} \left\{ F_{\text{even}}(u) + F_{\text{odd}}(u) W_{2K}^u \right\}$$

Using it to get K terms,

$$F(u+K) = \frac{1}{2} \left\{ F_{\text{even}}(u) - F_{\text{odd}}(u) W_{2K}^u \right\}$$

* Properties:

- FFT has time complexity of $O(N \log_2 N)$.
- FFT exhibits symmetry in its application.
- It can be implemented in place where input sequence is modified during computation.

* Applications:

- Fast large integer & polynomial multiplication.
- Filtering algorithms.
- Solving difference equations.
- Efficient matrix-vector multiplication.